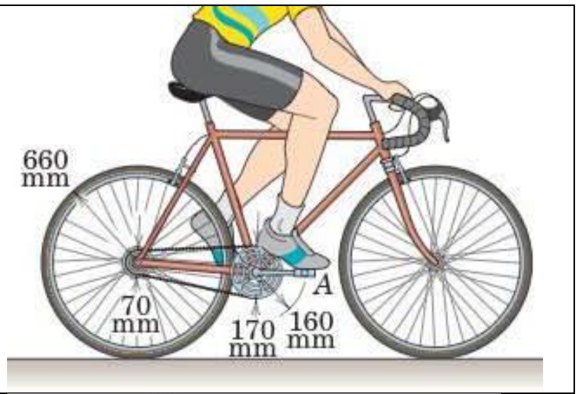
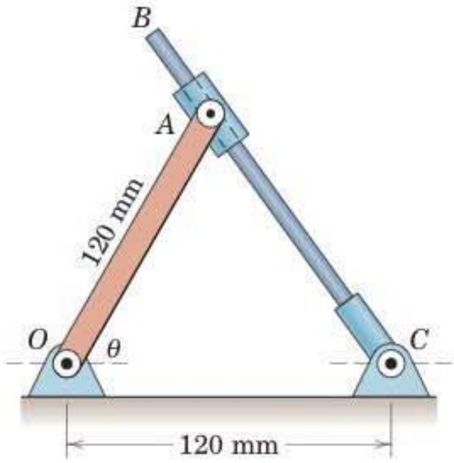
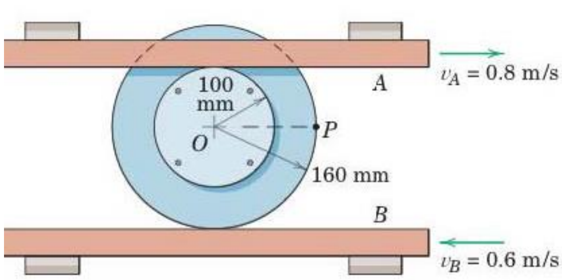
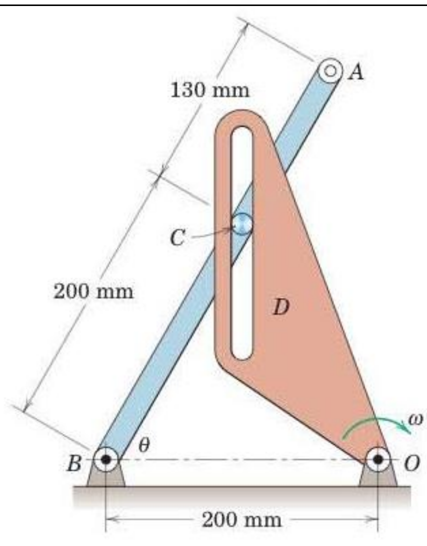
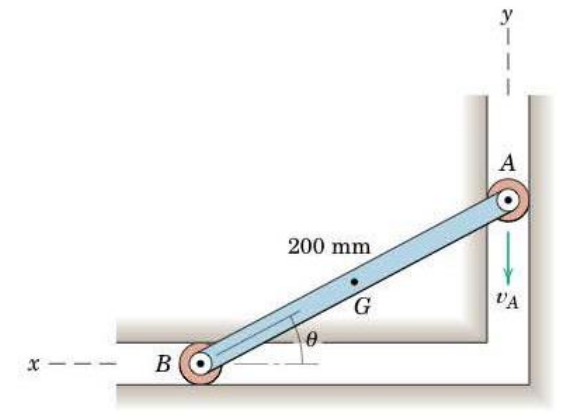
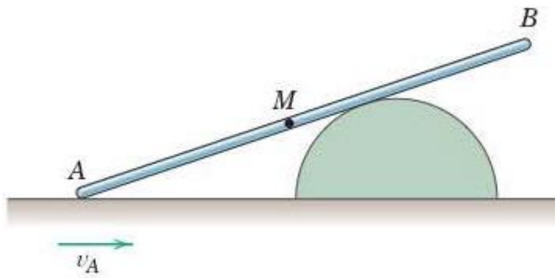
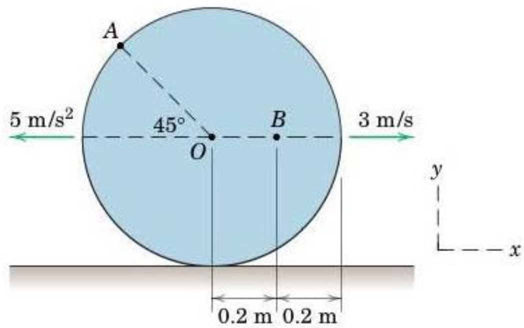
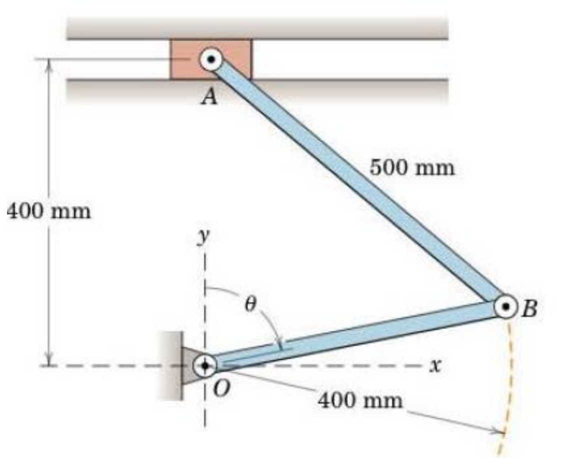
MEL1010 Engineering Mechanics
Tutorial 8: Relative Velocity, Instant Center of Zero Velocity and Relative Acceleration

Answer all of the following questions. Mention valid assumptions wherever applicable
1-4: Relative velocity, 5-6: Instant Center of Zero Velocity, 7-9: Relative Acceleration

1	The rider of the bicycle shown pumps steadily to maintain a constant speed of 16 km/h against a slight head wind. Calculate the maximum and minimum magnitudes of the absolute velocity of the pedal *A*. *Ans.* $(v_A)_{max} = 5.3$ m/s, . $(v_A)_{min} = 3.56$ m/s	
2	Rod *CB* slides through the pivoted collar attached to link *OA*. If *CB* has a clockwise angular velocity of 2 rad/s, determine the angular velocity ω_{OA} of link *OA* when θ = 60°. *Ans.* $\omega_{OA} = 4$ rad/s CW	
3	Each of the sliding bars *A* and *B* engages its respective rim of the two riveted wheels without slipping. Determine the magnitude of the velocity of point *P* for the position shown. *Ans.* $v_P = 0.9$ m/s	
4	For the instant represented the rotating link *D* has an angular velocity ω = 2 rad/s, and its slot is vertical. Also θ = 60° momentarily. Determine the velocity of end *A* of link *AB* for this instant. *Ans.* $v_A = 660$ mm/s	

5	End A of the link has a downward velocity v_A of 2 m/s during an interval of its motion. For the position where θ = 30°, determine the angular velocity ω of AB and the velocity v_G of the midpoint G of the link. (Use Instant Center of Zero Velocity Method) *Ans.* $\omega = 11.55\ \mathrm{rad/s\ CW},\ v_G = 1.155\ \mathrm{m/s}$	
6	End A of the slender pole is given a velocity v_A to the right along the horizontal surface. Show that the magnitude of the velocity of end B equals v_A when the midpoint M of the pole comes in contact with the semicircular obstruction. (Use Instant Center of Zero Velocity Method)	
7	The center O of the disk has the velocity and acceleration shown in the figure. If the disk rolls without slipping on the horizontal surface, determine the velocity of A and the acceleration of B for the instant represented. $\boldsymbol{v}_A = 5.12\,\boldsymbol{i} + 2.12\boldsymbol{j}$ m/s $\boldsymbol{a}_B = -16.25\,\boldsymbol{i} + 2.5\boldsymbol{j}$ m/s^2	
8	Determine the angular acceleration of link AB and the linear acceleration of A for θ = 90° if $\dot{\theta}$ = 0 and $\ddot{\theta}$ = 3 rad/s^2 at that position. Carry out your solution using vector rotation. *Ans.* $\boldsymbol{\alpha}_{AB} = -4\boldsymbol{k}\ \mathrm{rad/s^2},\ \boldsymbol{a}_A = 1.6\,\boldsymbol{i}\ \mathrm{m/s^2}$	
9	Link OA has a constant counterclockwise angular velocity ω during a short interval of its motion. For the position shown determine the angular accelerations of AB and BC. *Ans.* $\alpha_{AB} = 2\omega^2, \alpha_{BC} = 0$